Expansion of deforestation in the Cerrado of Maranhão, Brazil, 2001–2023

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> This report is built from `report_en.ipynb`, which re-runs the analysis notebooks `analise_municipios.ipynb`, `analise_microregioes.ipynb` and `analise_grade_micro.ipynb` on the data in the repository github.com/LambdaGeo/analise-cerrado-ma. Every number, table and figure is computed when the report is built. It summarizes the peer-reviewed article, published in Portuguese: doi:10.70860/rtg.v15i35.19356. Please cite the article.

Abstract. The Cerrado covers about two thirds of the state of Maranhão, and in recent years the state has become one of the main fronts of deforestation in Brazil. Using PRODES Cerrado data from INPE, we examined the absolute and relative evolution of deforestation in the Cerrado of Maranhão from 2001 to 2023, by biome, microregion and municipality. By 2023, 89,880 km² had been deforested, 41% of the Cerrado in the state, and 54% of that area was cleared from 2001 onwards. Annual clearing fell after 2008 and stayed between about 1,300 and 1,900 km² a year from 2011 to 2019, then rose every year from 2020, reaching 2,928 km² in 2023. Deforestation spread over the whole biome rather than only the south of the state: on a regular grid, which removes the effect of the size of administrative units, the share of cells with less than 20% of their area deforested fell from 72% in 2000 to 30% in 2023. Presidente Dutra, Caxias and Chapadinha were the microregions with the largest increase in deforested share; in Presidente Dutra it went from 44% to 82%. Among municipalities, Balsas had the largest deforestation from 2001 to 2023 (3,581 km²), while in Governador Eugênio Barros the deforested share of the municipality rose from 26% to 89%. The 15 municipalities with the largest relative increase cover an area equal to 8% of the Cerrado in the state but account for 14% of its deforestation since 2001.

Keywords: deforestation; Cerrado; PRODES; Maranhão; remote sensing.

1. Introduction

The Cerrado is the Brazilian savanna, a biodiversity hotspot of about two million square kilometres whose conversion to agriculture accelerated in recent decades (Strassburg et al., 2017). Clearing its deep-rooted vegetation changes carbon storage and the water cycle (Spera et al., 2016). Unlike rural properties in the Legal Amazon, which must keep 80% of their native vegetation, properties in the Cerrado need to keep only 20% (Brasil, 2012), so much of the clearing is legal.

In Maranhão the Cerrado covers about two thirds of the state. In 2023 the state rose from fifth to first place in the national deforestation ranking, and it has 13 of the 50 municipalities with the most deforestation in the last four years (MapBiomas, 2024). Reports and studies tend to focus on the large municipalities of the south of the state. This study looks at the whole Cerrado of Maranhão over more than two decades, in absolute and relative terms, to see where deforestation has been spreading.

2. Data and methods

Deforestation data. We used PRODES Cerrado (INPE), which maps the removal of native vegetation regardless of its later use or legality, as published on the TerraBrasilis platform (data downloaded in May 2024). PRODES gives the deforestation accumulated up to 2000, then increments every two years until 2012 and every year from 2013. For the biennial period, each year was given half of the two-year increment.

Territorial units. Deforestation was summarized for the whole Cerrado of Maranhão (217,029 km²), for the 14 IBGE microregions with more than half of their territory in the biome, for the 139 municipalities covered by PRODES Cerrado, and for a regular grid of 2,805 cells of 0.083° (about 9 km on a side), which removes the influence of unit size. For microregions the share is relative to the Cerrado area inside each microregion, for municipalities to the whole municipal area, and for grid cells to the cell area.

Measures. For each unit we computed the deforestation up to 2000, from 2001 to 2023, and the total up to 2023, both in km² (absolute) and as a share of the unit's area (relative).

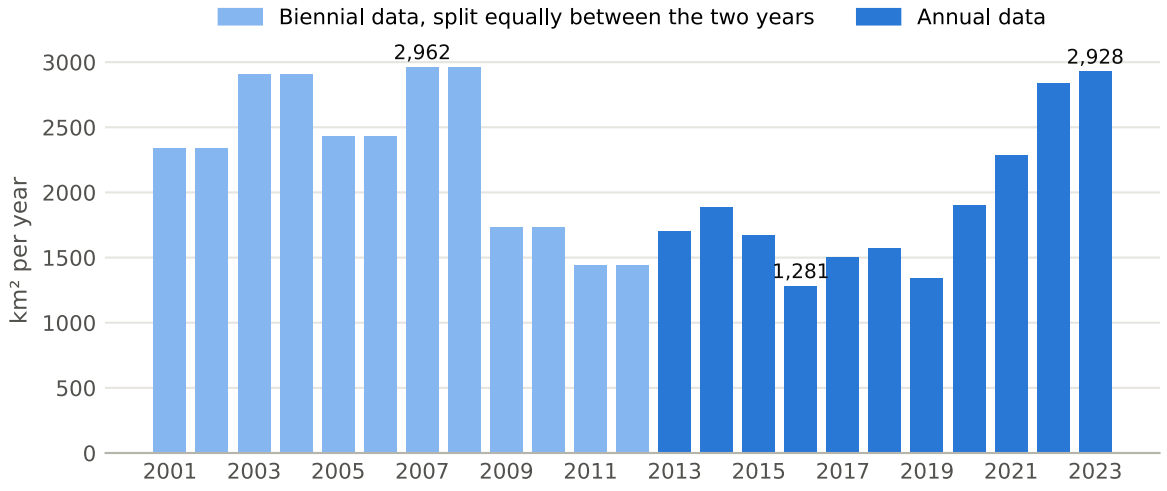
3. Results

3.1 The biome

By 2023, 89,880 km² of the Cerrado of Maranhão had been deforested, 41% of its area. Of this, 41,357 km² had been cleared by 2000 and 48,523 km² from 2001 to 2023, so more than half of all deforestation in the biome happened in the period studied.

Annual clearing peaked at 2,962 km² in 2007–2008 (Figure 1). It then fell to 1,440 km² a year in 2011–2012 and stayed between about 1,300 and 1,900 km² until 2019, with the lowest value in 2016 (1,281 km²). From 2020 it rose every year, reaching 2,928 km² in 2023, almost back to the peak.

Figure 1. Annual deforestation in the Cerrado of Maranhão, 2001–2023 (PRODES).



3.2 Microregions

Deforestation grew in every microregion (Table 1). The largest increases in deforested share were in the east of the state: Presidente Dutra, where it went from 44% to 82% of the Cerrado area, Caxias and Chapadinha. Gerais de Balsas, the largest microregion, had the most deforestation in km² from 2001 to 2023 (about 8,000 km²), but ranks only

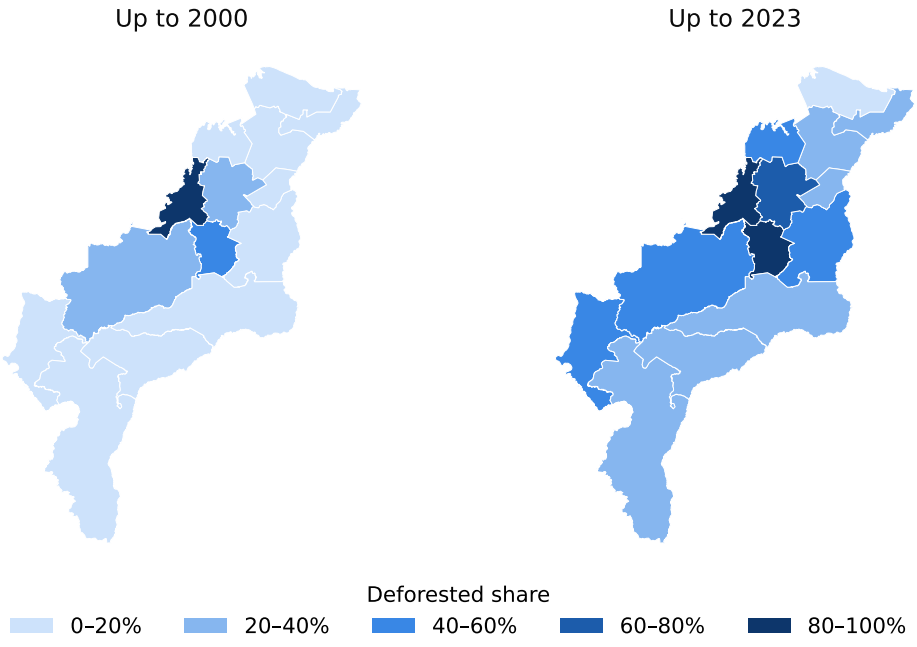
tenth in relative terms. The smallest increases were in Médio Mearim, which was already 93% deforested in 2000, and in Lençóis Maranhenses, home to the national park of the same name. Figure 2 shows the shift: in 2000, 10 of the 14 microregions had less than 20% of their Cerrado cleared; by 2023 half of them had more than 40%, and only Lençóis Maranhenses remained below 20%.

Table 1. Deforestation by microregion, ordered by the increase in deforested share.

		Deforested share of the Cerrado area			
Microregion	Cerrado area (km²)	2000	2023	Increase (p.p.)	Deforested 2001–2023 (km²)
Presidente Dutra	6,544	44%	82%	37	2,447
Caxias	15,340	10%	43%	32	4,934
Chapadinha	10,799	4%	32%	28	3,007
Coelho Neto	3,608	7%	34%	27	982
Codó	9,915	36%	62%	26	2,550
Baixo Parnaíba Maranhense	6,040	9%	35%	25	1,534
Itapecuru Mirim	6,609	19%	45%	25	1,673
Porto Franco	14,062	18%	43%	25	3,498
Alto Mearim e Grajaú	33,701	23%	46%	23	7,811
Gerais de Balsas	36,657	10%	31%	22	8,003
Chapadas das Mangabeiras	16,961	9%	30%	21	3,583
Chapadas do Alto Itapecuru	25,011	6%	24%	19	4,665
Lençóis Maranhenses	9,900	3%	14%	11	1,114
Médio Mearim	6,887	93%	97%	4	284

p.p.: percentage points. Source: PRODES Cerrado (INPE); IBGE 2022 microregions.

Figure 2. Share of the Cerrado area of each microregion deforested up to 2000 and up to 2023.



3.3 Municipalities

In absolute terms, deforestation is led by the large municipalities of the south (Table 2). Balsas had the most deforestation both in total (5,293 km² by 2023) and from 2001 to 2023 (3,581 km²), followed by Grajaú. Carolina had the largest relative growth among these municipalities: its deforested area went from 273 km² in 2000 to 1,566 km² in 2023, almost six times more. The 15 municipalities in Table 2 account for 43% of the deforestation from 2001 to 2023.

Table 2. The 15 municipalities with the most deforestation up to 2023 (article Table 3).

			Deforested (km²)			
Rank	Municipality	Area (km²)	Up to 2000	Up to 2023	2001–2023	Growth since 2000
1	Balsas	13,141	1,712	5,293	3,581	209%
2	Grajaú	8,862	1,213	3,783	2,569	212%
3	Barra do Corda	5,188	2,081	3,079	998	48%
4	Codó	4,362	1,459	2,637	1,178	81%
5	Tuntum	3,369	1,183	2,190	1,007	85%
6	Riachão	6,403	625	2,161	1,536	246%
7	Amarante do Maranhão	7,440	1,079	1,927	848	79%
8	Parnarama	3,246	529	1,809	1,280	242%
9	Alto Parnaíba	11,127	510	1,809	1,299	255%
10	Caxias	5,202	405	1,614	1,209	299%
11	Tasso Fragoso	4,369	477	1,575	1,098	230%
12	Carolina	6,268	273	1,566	1,292	473%
13	Sítio Novo	3,115	404	1,483	1,080	268%
14	Formosa da Serra Negra	3,691	345	1,418	1,073	311%
15	Coroatá	2,264	715	1,367	652	91%
	Total	**88,044**	**13,011**	**33,712**	**20,701**	

Growth since 2000: deforestation from 2001 to 2023 divided by deforestation up to 2000. These 15 municipalities cover 40.6% of the Cerrado area of the state and account for 42.7% of its deforestation from 2001 to 2023.

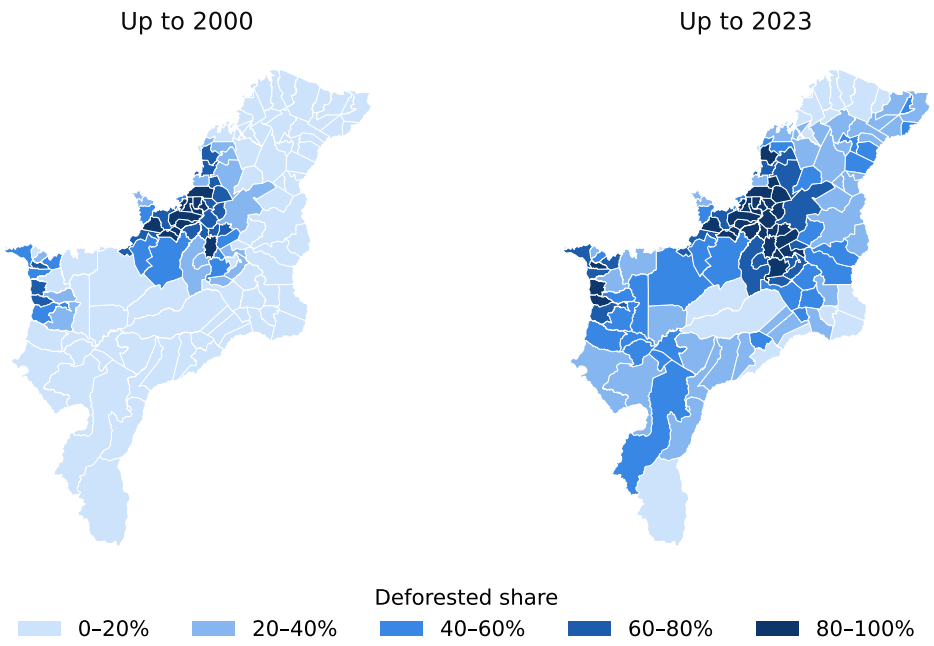
Relative measures tell a different story (Table 3). The municipalities with the largest increase in deforested share are mostly small and lie in the east of the state, outside the rankings that usually draw attention. In Governador Eugênio Barros the deforested share went from 26% of the municipality in 2000 to 89% in 2023. Together, these 15 municipalities cover 16,539 km², 8% of the Cerrado in the state, but account for 6,911 km², 14% of the deforestation from 2001 to 2023. Figure 3 shows the same pattern: by 2023 deforestation had reached high shares across the east and centre of the biome, not only in the south.

Table 3. The 15 municipalities with the largest increase in deforested share, 2001–2023 (article Table 4).

			Deforested share of the municipality			
Rank	Municipality	Area (km²)	2000	2023	Increase	Deforested 2001–2023 (km²)
1	Governador Eugênio Barros	648	26%	89%	63%	407
2	Governador Luiz Rocha	402	16%	72%	56%	225
3	Senador Alexandre Costa	426	18%	71%	53%	227
4	São Domingos do Maranhão	1,152	41%	85%	44%	503
5	Anapurus	609	5%	48%	44%	265
6	Graça Aranha	271	41%	84%	43%	117
7	São João do Soter	1,438	7%	51%	43%	621
8	Matões	2,109	9%	52%	43%	899
9	Santa Filomena do Maranhão	623	24%	64%	40%	249
10	Fortuna	835	21%	61%	40%	331
11	Parnarama	3,246	16%	56%	39%	1,280
12	Água Doce do Maranhão	442	14%	53%	38%	169
13	Gonçalves Dias	884	43%	81%	38%	334
14	Buriti	1,476	6%	44%	38%	557
15	Colinas	1,979	12%	49%	37%	727

Increase: deforestation from 2001 to 2023 as a share of the municipal area, in percentage points.

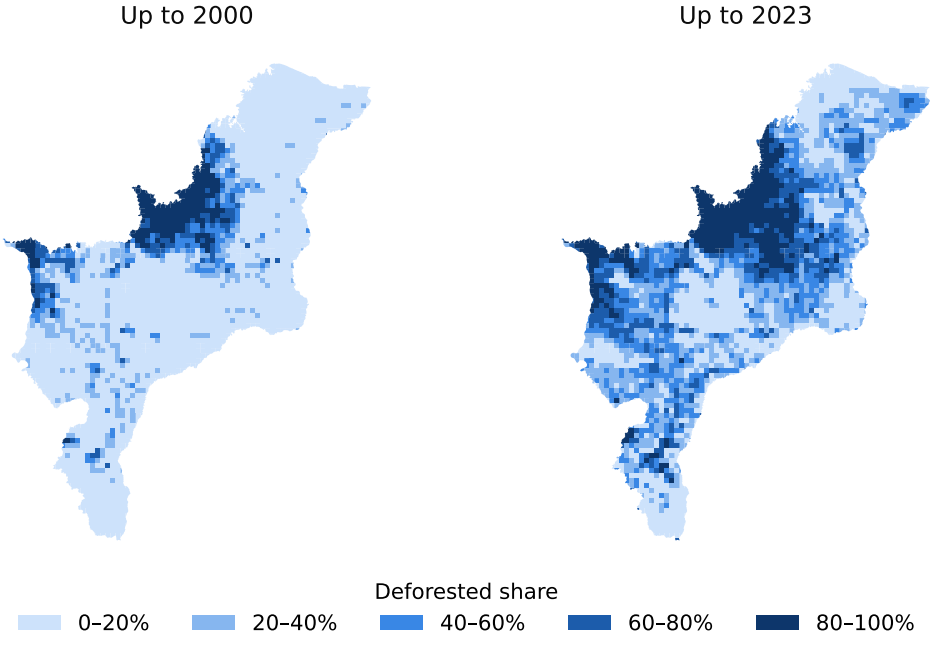
Figure 3. Share of each municipality deforested up to 2000 and up to 2023, within the Cerrado of Maranhão.



3.4 Regular grid

The grid confirms the spread without the bias of unit size (Figure 4). In 2000, 72% of the cells (2,016 of 2,805) had less than 20% of their area deforested, and the cells cleared above 80% were concentrated in the north of the biome, around Médio Mearim and Pindaré, and along its western edge. By 2023 only 30% of the cells (855) remained below 20%, and the number of cells more than 80% deforested had almost doubled, from 215 to 407. Cells with more than 40% cleared now appear from the north to the east and the south of the biome; the main areas still mostly below 20% are a block in its centre, the far north-east and the southern tip.

Figure 4. Share of each grid cell (0.083°, about 9 km) deforested up to 2000 and up to 2023 (article Figure 7).



4. Discussion and limitations

Where deforestation is happening. Absolute figures point to the large municipalities of the south, such as Balsas and Grajaú, which also rank third and ninth in deforestation from 2001 to 2023 in the whole Brazilian Cerrado (article Table 2, from TerraBrasilis). Relative figures show that clearing has also advanced quickly in the east, in microregions such as Presidente Dutra, Caxias and Chapadinha and in small municipalities that are close to losing all their native vegetation. Both views are needed: absolute figures favour large units and relative figures favour small ones.

Trend. The rise since 2020 brings annual clearing back to the levels of the late 2000s. Explaining it is beyond the scope of this exploratory study; the legal framework, which requires only 20% of native vegetation to be kept on Cerrado properties, is one factor discussed in the article.

Units of analysis. Municipal shares are relative to the whole municipal area, including any part outside the Cerrado, while microregion shares are relative to the Cerrado area inside each microregion. The grid map of deforestation from 2015 to 2023 (article Figure 8) and the deforestation in Indigenous Lands (article Table 5, from TerraBrasilis) are not reproduced here.

Biennial data. Before 2013 PRODES Cerrado is biennial, and splitting each increment equally between two years hides any difference between them.

Notes on the published figures. The article's text gives 21,139 km² for the deforestation of "these 15 municipalities"; that sum belongs to the ranking by deforestation from 2001 to 2023, which shares 13 municipalities with the list in Table 2 (whose sum is 20,701 km²). The conclusion is the same. The analysis uses 217,029 km² for the Cerrado of Maranhão.

5. Conclusions

From 2001 to 2023 the Cerrado of Maranhão lost 48,523 km² of native vegetation, more than everything cleared before 2000, and by 2023 41% of the biome in the state had been deforested. After a decline in the 2010s, clearing has risen every year since 2020. Deforestation is not limited to the south of the state or to large municipalities: on the regular grid, cells with less than 20% of their area cleared went from 72% to 30% of the biome; in relative terms the fastest losses are in the east, and the 15 municipalities with the largest relative increase account for 14% of the deforestation since 2001 while covering 8% of the biome. Future work should look at how the cleared land is being used, for example with MapBiomas land use and land cover maps.

Reproducibility

Code, data and notebooks are in github.com/LambdaGeo/analise-cerrado-ma (MIT license for the code; the data derive from INPE and IBGE and follow their terms of use). With Python 3.12 and the pinned dependencies in `requirements.txt`, the notebooks `analise_bioma.ipynb`, `analise_microregioes.ipynb`, `analise_municipios.ipynb` and `analise_grade_micro.ipynb` reproduce the published results, and `python build_report.py` rebuilds this report. The report checks the values quoted in its text against the analysis and stops if they differ.

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